

# Modeling Conditional Variance: ARCH and GARCH Model

- **Heteroskedasticity**: the variance changes over time (as opposite to homoskedasticity)
  - Periods of unusually large volatility is followed by periods of relative tranquility.
  - Stock market returns determines the riskiness of our investment.
- When a process is heteroskedastic, it will follow ticker-tailed probability distributions, due to the volatility clustering: we can compute the kurtosis of the variable to check it (the kurtosis of a normal distribution is equal to 3)

# Unconditional and Conditional Variance

- **Long-run variance**: the unconditional variance is not useful if you are interested in forecast a variance of an asset over the holding period.
- **Conditional forecasts** are better because they take into account the known current and past realizations of the series (if you are assuming rational expectations, then the economic agent use past, current and future information)

# Unconditional Variance

$$Y_t = Y_{t-1} + e_t$$
$$Y_t = Y_o + \sum_{i=1}^t e_t$$

With  $E[e_t] = 0$ ; and  $Var[e_t] = \sigma^2$  (*constant*)

The unconditional  $Var(Y_t)$  is *unbounded*:

$$Var(Y_t) = Var(Y_o + \sum_{i=1}^t e_t)$$
$$Var(Y_t) = t\sigma^2$$

# Conditional Variance

$$Y_t = Y_{t-1} + e_t$$

$$Y_t - Y_{t-1} = e_t$$

$$E(Y_t|Y_{t-1}) = E(Y_{t-1}|Y_{t-1}) + E(e_t)$$

$$E(Y_t|Y_{t-1}) = Y_{t-1}$$

$$Var(Y_t|Y_{t-1}) = E[Y_t - E(Y_t|Y_{t-1})]^2 = E(e_t^2)$$

$$Var(Y_t|Y_{t-1}) = Var(e_t) = \sigma^2$$

Then conditional variance is constant.

# Conditional Variance

What if  $Var(e_t) = \sigma_t^2$  ?

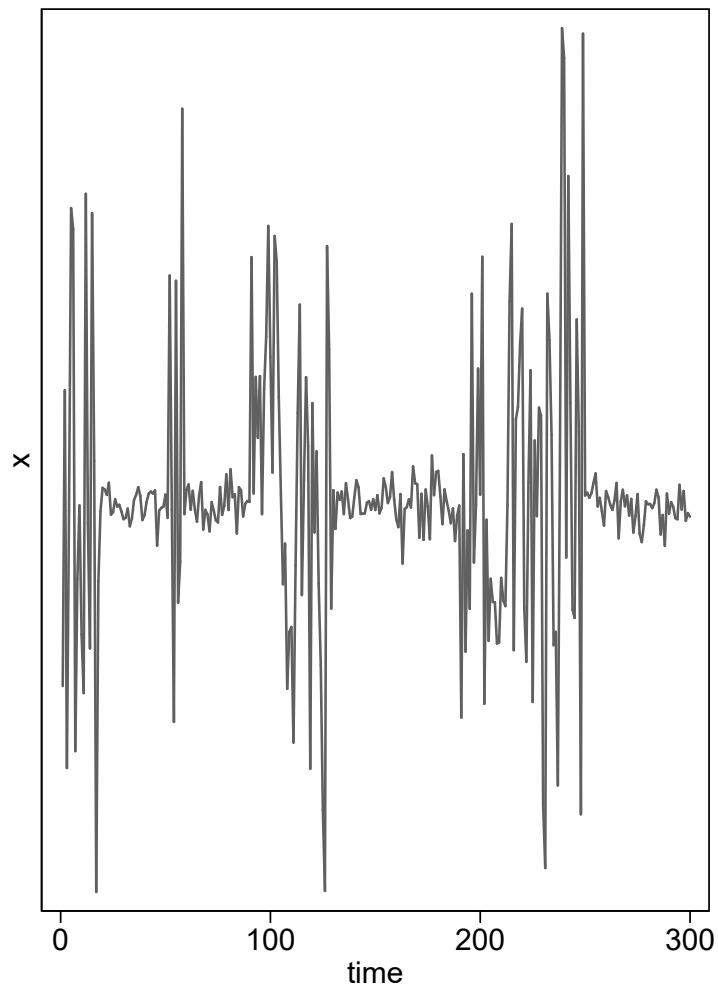
Then conditional variance will not be constant.

We could estimate the conditional variance as an AR(p)  $\Rightarrow$  Autoregressive Condition Heteroskedasticity Model (ARCH model).

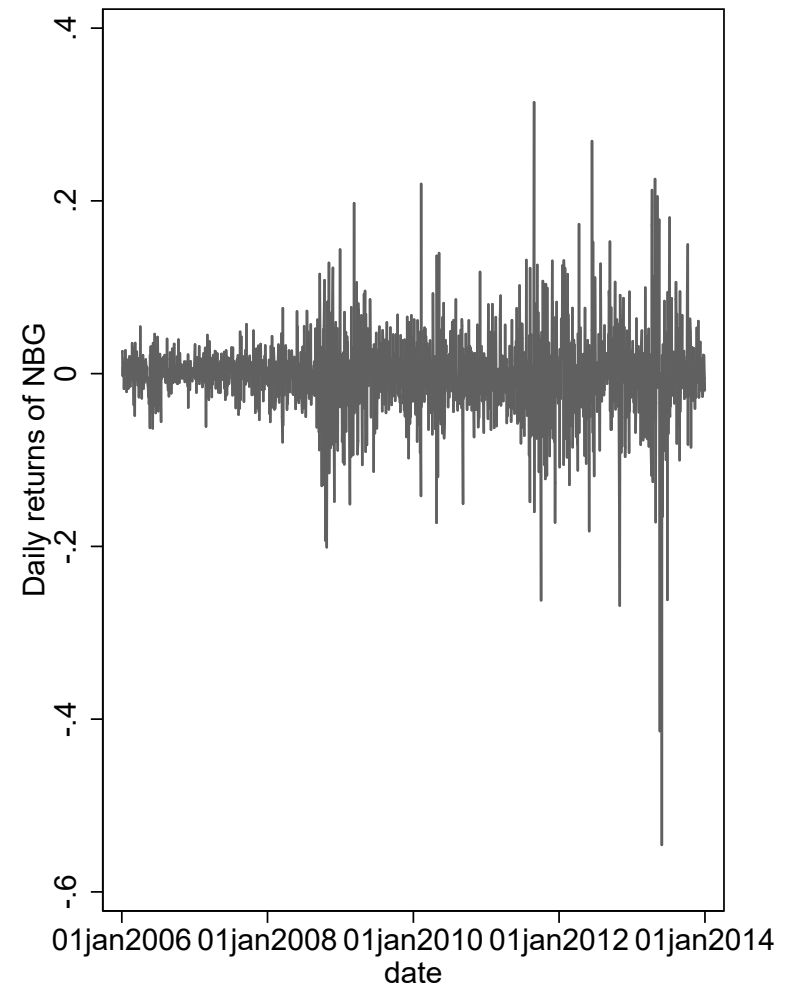
OBS: the residuals of the model can come from and ARMA model of even a standard regression model.

- \* The model for the mean and the conditional variance should be estimated simultaneously (By Maximum Likelihood) and the error from the AR(p) model on the variance is treated as a multiplicative disturbance instead of additive (for better results).

Artificial Data



National Bank of Greece returns



# ARCH(1) MODEL

- Engle(1982) proposed a model where the conditional variance of process could be modeled as a linear combination of past squared errors.
- Engle proposed the following multiplicative conditionally heteroskedasticity:

$$e_t = v_t \sqrt{\alpha_0 + \alpha_1 e_{t-1}^2}$$
$$v_t \sim iid(0,1)$$

\*  $v_t$  and  $e_t$  are independent random variables.

The unconditional mean is:

$$E(e_t) = E(v_t) * E(\alpha_0 + \alpha_1 e_{t-1}^2)^{1/2} = 0$$

# Unconditional Variance

What is the unconditional  $\text{Var}(e_t)$ ?

$$\text{Var}(e_t) = E[e_t^2] = E\left(v_t^2 * (\alpha_0 + \alpha_1 e_{t-1}^2)\right)$$

$$\text{Var}(e_t) = E(v_t^2) * E(\alpha_0 + \alpha_1 e_{t-1}^2)$$

$$\text{Var}(e_t) = 1 * E(\alpha_0 + \alpha_1 e_{t-1}^2)$$

Because  $0 < \alpha_1 < 1$ ,  $E(e_t^2) = E(e_{t-1}^2) = \text{Var}(e_t)$  (stationary process)

Then:

$$\text{Var}(e_t) = \alpha_0 + \alpha_1 \text{Var}(e_t)$$

$$\text{Var}(e_t) = \frac{\alpha_0}{1 - \alpha_1}$$

The unconditional mean and variance are unaffected by the error process and has a constant variance.



# Conditional Variance

- The conditional mean of  $e_t$ :

$$E(e_t|e_{t-1}) = E(v_t) * (\alpha_0 + \alpha_1 e_{t-1}^2)^{1/2} = 0$$

- The conditional variance of  $e_t$ :

$$\begin{aligned} Var(e_t|e_{t-1}) &= E(e_t^2|e_{t-1}) - (E(e_t|e_{t-1}))^2 \\ Var(e_t|e_{t-1}) &= E[(\alpha_0 * v_t^2 + \alpha_1 e_{t-1}^2 * v_t^2)|e_{t-1}] \\ Var(e_t|e_{t-1}) &= \alpha_0 + \alpha_1 e_{t-1}^2 \end{aligned}$$

\* The conditional variance of  $Y_t$  exhibits time-varying volatility ( $e_t$  follows an AR(1) : ARCH model); the errors are conditionally heteroskedastic.

# ARCH(P)

- The ARCH process can be extended as and ARCH(p):
- Engle proposed the following multiplicative conditionally heteroskedasticity:

$$e_t = v_t \sqrt{\alpha_0 + \sum_{i=1}^p \alpha_i e_{t-i}^2}$$

- The conditional variance acts like an autoregressive process of order q.
- The conditional variance will revert to unconditional variance provided (because  $0 < \alpha_1 < 1$ , )

# TESTING FOR ARCH(q)

- We can test for autocorrelation in the squared residuals by examining the ACF.
- **Method 1: Ljung-Box test:**
  - **First:** estimate your model for the desired time series and get the residual; check if the residuals follows a white noise process. The rejection of it is an indication of ARCH.
- **STATA COMMAND:**

wntestq  $e^2$ , lags(4)

(we can also plot the autocorrelation of the squared residuals and the statistic Q is provided for each lag).

# TESTING FOR ARCH( $q$ )

- **Method 2.** A test proposed by Engle (known as Engle LM test) based on Lagrange Multiplier ( LM ). The test consist on regressing the squared residuals (the residuals of the estimate mean equation) on a constant and on the  $p$  lagged residuals. If no ARCH effect, we should do not reject the all the coefficients are equal to zero (F- test for small sample,  $p$  df in the numerator and  $T-p$  df in the denominator; for large samples,  $TR^2$  converges to a  $\chi^2$  with  $p$  df); by rejecting the null, we have evidence that there is ARCH effects.

# How to choose the optimum lag length?

- We can estimate several models with different lags and then compare them using BIC, and AIC information criteria.
- **STATA Command:**

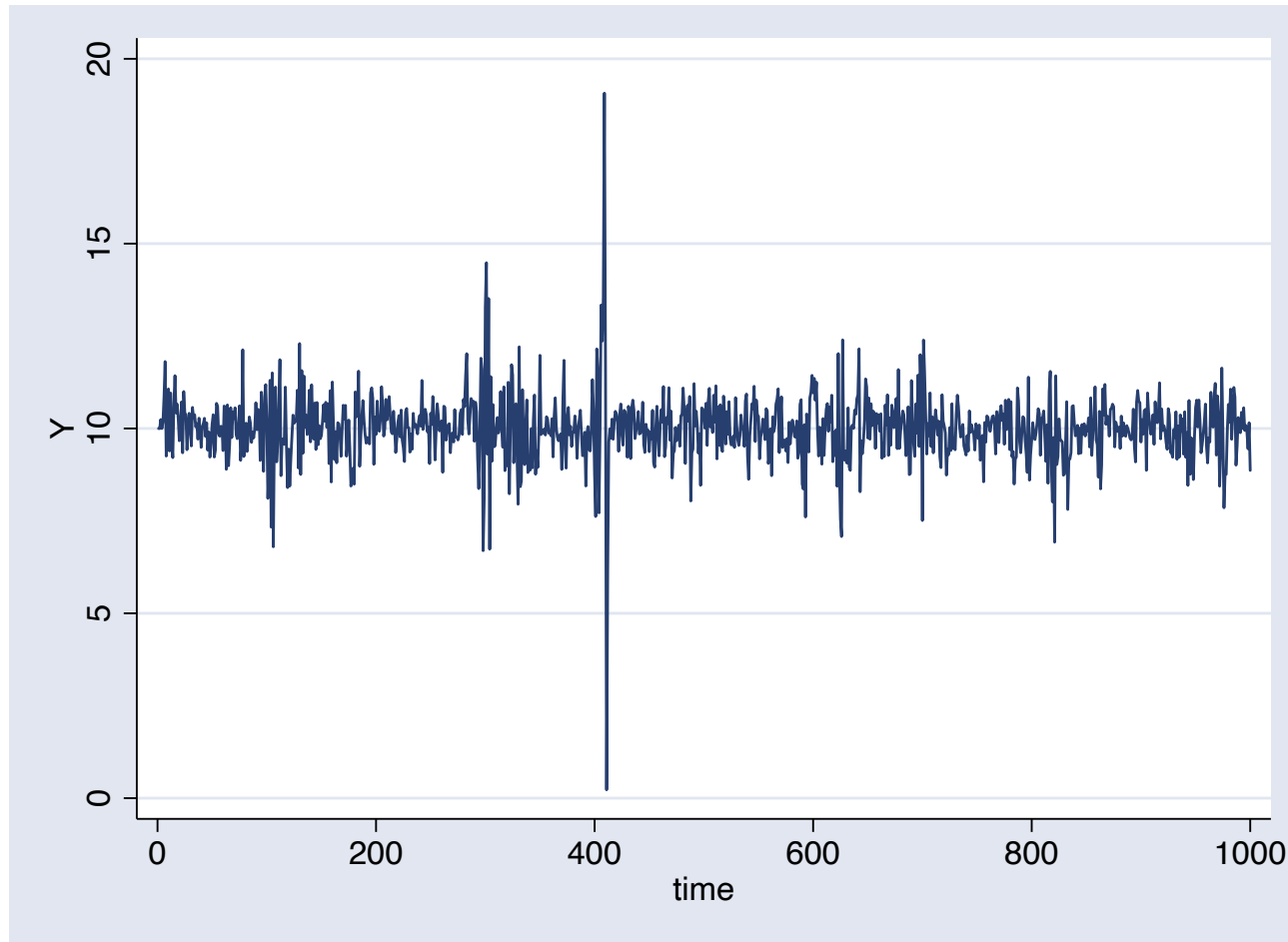
**estat ic**

# Simulating an ARCH(2)

```
drop _all
set obs 1000
set seed 12345
gen time = _n
tsset time
local beta0 = 10
local alpha0 = 0.20
local alpha1 = 0.30
local alpha2 = 0.40
gen u = rnormal(0,1)
gen e = .
replace e = 0 in 1/2
replace e = u*(`alpha0' + `alpha1'*(L.e^2) + `alpha2'*(L2.e^2) )^(1/2) in 3/L
gen Y = `beta0' + e
graph twoway line Y time
graph save AR-0-ARCH-2.gph, replace
graph export AR-0-ARCH-2.pdf, replace
qnorm Y
graph save qqAR-0-ARCH-2.gph, replace
graph export qqAR-0-ARCH-2.pdf, replace
histogram Y, normal
graph save histAR-0-ARCH-2.gph, replace
graph export histAR-0-ARCH-2.pdf, replace
tabstat Y, statistics(kurtosis)
arch Y, arch(1/2)
```

# Example: ARCH(2)

$$\text{Var}(e_t|e_{t-1}) = \alpha_0 + \alpha_1 e_{t-1}^2 + \alpha_2 e_{t-2}^2$$



# Diagnostic Checking: ARCH(2)

## Kurtosis (unconditional)

variable | kurtosis

-----+-----

Y | 25.82196

-----

- Kurtosis > 3 (indicates fat tail)

## Ljung-Box (Q) test

wntestq e2

Portmanteau test for white noise

-----

Portmanteau (Q) statistic = 355.6252

Prob > chi2(40) = 0.0000

## AIC/BIC lag-selection

```
forvalues lags=1/10{
```

```
display "Lags = " `lags'
```

```
quietly arch Y, arch(1/`lags')
```

```
estat ic
```

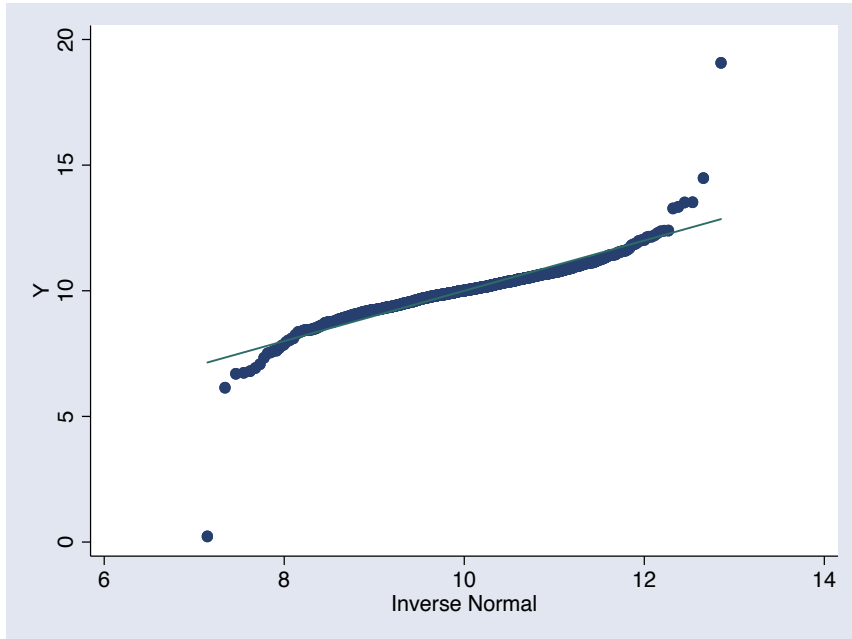
```
}
```

- AIC, BIC indicates two lags (as our DGP is and ARCH(2))

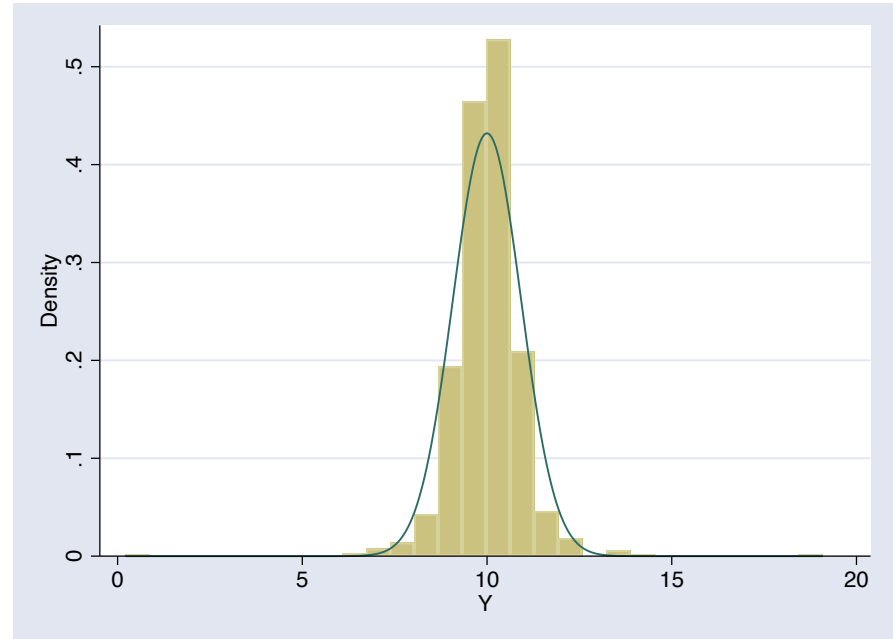


# Diagnostic Checking: ARCH(2)

Q-Q Plot



Histogram



QQ-Plot show a distribution that is skewed (more fat tail);  
Histogram also is not symmetrical.

# Estimating ARCH(2)

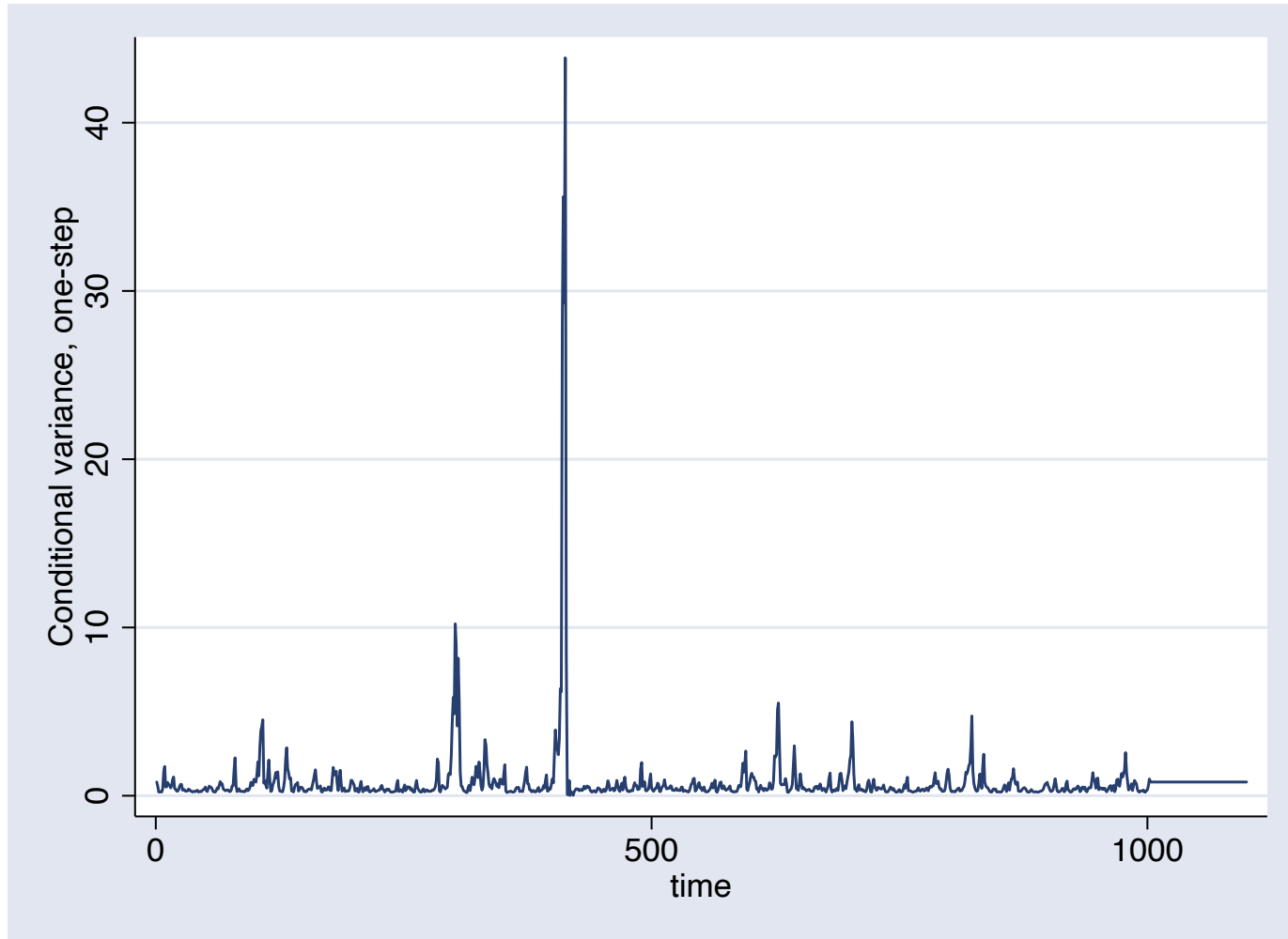
```

-----
      Y |   Coef.  Std. Err.   z  P>|z|   [95% Conf. Interval]
-----+-----
Y      |
  _cons |  10.00629   .0192137  520.79  0.000   9.968632   10.04395
-----+-----
ARCH   |
  arch  |
    L1. |   .288942   .0476651   6.06  0.000   .19552   .3823639
    L2. |   .424621   .0623969   6.81  0.000   .3023253   .5469167
      |
  _cons |   .21235   .0191894  11.07  0.000   .1747394   .2499606
-----

```

**Our DGP:**  $Y_t = 10 + e_t$ ;  $e_t = u * (0.2 + 0.3 * e_{t-1} + 0.40 * e_{t-2})^{1/2}$

# Predicting the Variance (it converges to the unconditional variance)



# PROBLEMS IN ARCH MODELING

- In order to ensure with nonnegativity and stationarity constraint, Engle imposed a declining set of weights on the errors (so a restricted ARCH); an unrestricted ARCH would involve too many lags and could lead to negative variance and non-stationarity problems.

# GARCH MODEL(1,1)

- Bollerslev (1986) extended Engle's work to allow for the conditional variance to be an ARMA process, allowing for AR and MA components in the heteroskedastic variance (a more parsimonious model). Example GARCH(1,1):

$$Var(e_t|e_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 e_{t-1}^2 + \beta \sigma_{t-1}^2$$

(by recursively substituting the value of  $\sigma_t^2$ , this would be and ARCH( $\infty$ ))

# GARCH MODEL(p,q)

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i e_{t-i}^2 + \sum_{i=1}^q \beta_i \sigma_{t-i}^2$$

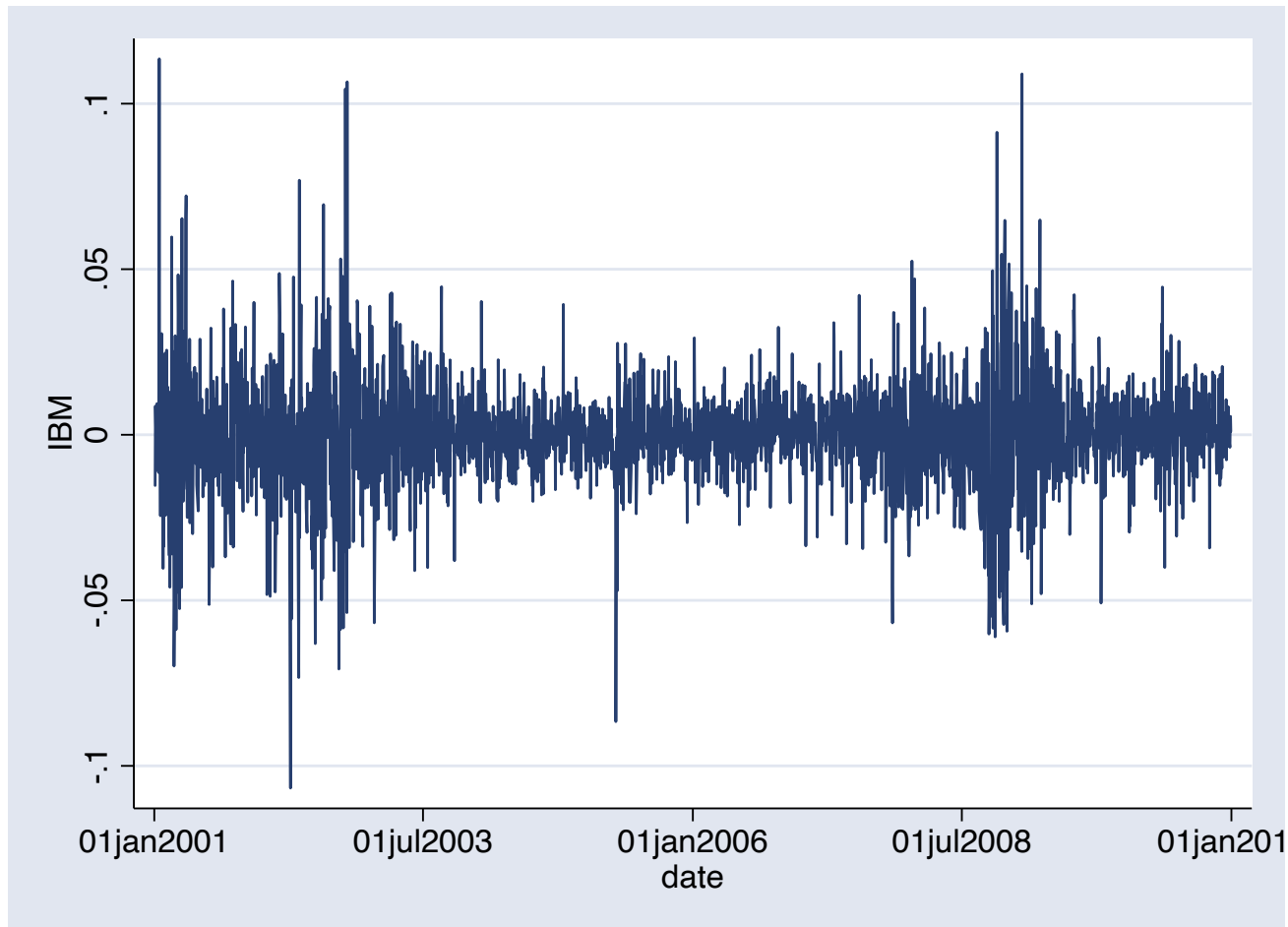
- In practice  $p \leq 2$  and  $q \leq 2$
- For the conditional variance to be stationary:

$$\sum_{i=1}^p \alpha_i + \sum_{i=1}^q \beta_i < 1$$

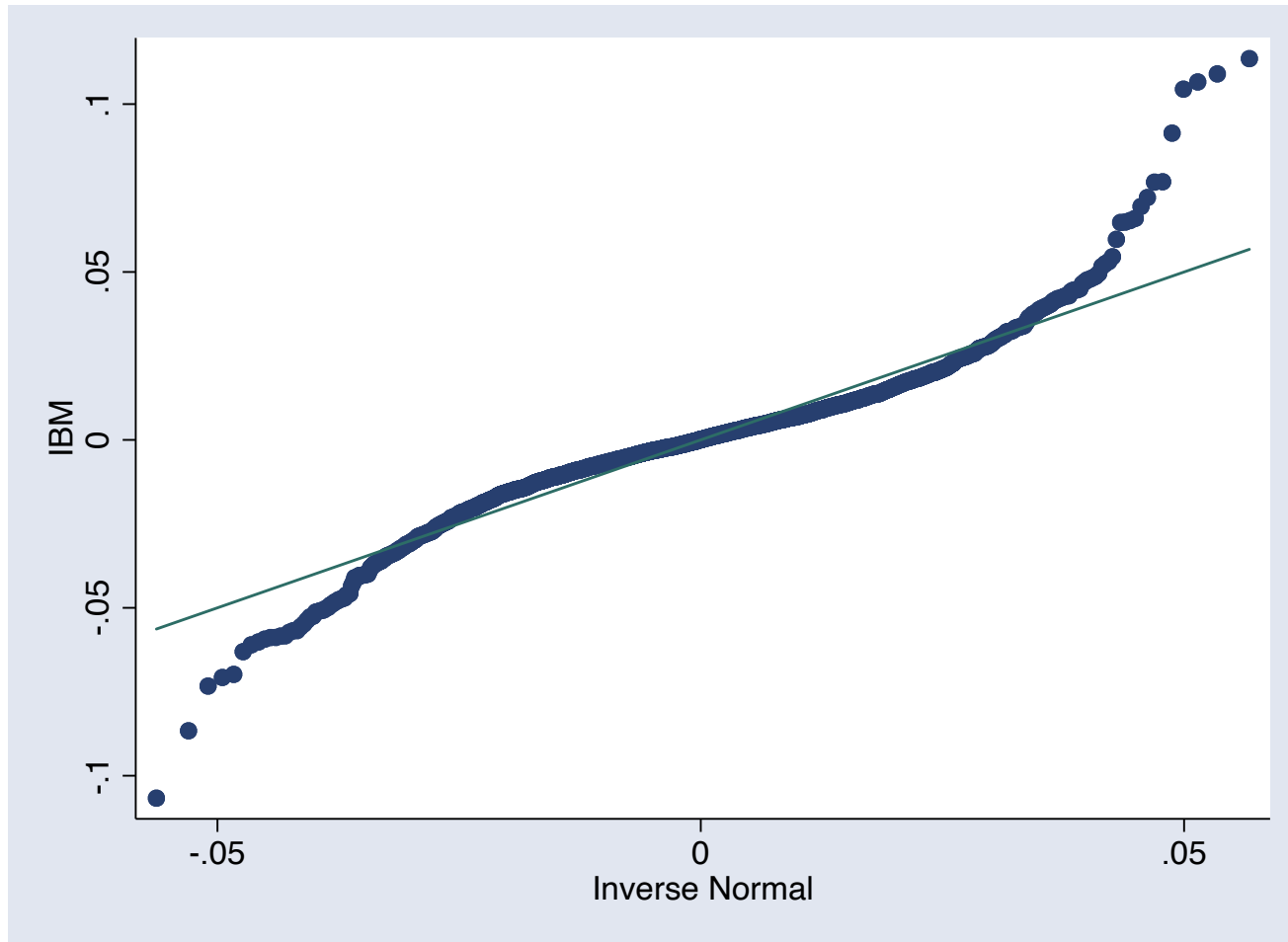
In a GARCH(1,1):  $\alpha_1 + \beta < 1$  (if it is equal to 1  $\Rightarrow$  Integrated GARCH, or IGARCH)

# GARCH Example: IBM Data

Daily % daily returns: 2000-2010



# QQ-Plot





# Ljung-Box test

wntestq e2

Portmanteau test for white noise

-----

Portmanteau (Q) statistic = 1604.3685

Prob > chi2(40) = 0.0000

\*the Ljung-Box test indicate volatility clustering: ARCH or GARCH model

## Lag Selection for GARCH(p,q) model: AIC and BIC Information Criteria

| ARCH(p) | GARCH(q) | AIC              | BIC              |
|---------|----------|------------------|------------------|
| 1       | 0        | -13715.52        | -13698.04        |
| 1       | 1        | -14236.17        | -14212.85        |
| 1       | 2        | -14238.1         | -14208.95        |
| 2       | 0        | -13901.84        | -13878.52        |
| 2       | 1        | <b>-14243.92</b> | <b>-14214.78</b> |
| 2       | 2        | -14237.75        | -14202.78        |

Both criteria suggest an ARCH(2,1)

# Estimating the model

| -----       |       |           |          |        |                      |           |           |
|-------------|-------|-----------|----------|--------|----------------------|-----------|-----------|
| IBM         | Coef. | Std. Err. | z        | P> z   | [95% Conf. Interval] |           |           |
| -----+----- |       |           |          |        |                      |           |           |
| IBM         |       |           |          |        |                      |           |           |
| _cons       |       | .0005631  | .0002232 | 2.52   | 0.012                | .0001257  | .0010006  |
| -----+----- |       |           |          |        |                      |           |           |
| ARCH        |       |           |          |        |                      |           |           |
| arch        |       |           |          |        |                      |           |           |
| L1.         |       | .1867001  | .0145153 | 12.86  | 0.000                | .1582506  | .2151496  |
| L2.         |       | -.1003556 | .0155189 | -6.47  | 0.000                | -.1307721 | -.0699391 |
|             |       |           |          |        |                      |           |           |
| garch       |       |           |          |        |                      |           |           |
| L1.         |       | .9023334  | .0077119 | 117.01 | 0.000                | .8872184  | .9174484  |
|             |       |           |          |        |                      |           |           |
| _cons       |       | 3.63e-06  | 5.04e-07 | 7.19   | 0.000                | 2.64e-06  | 4.62e-06  |

# Ljung-Box test on the Standardized Squared Residual

Portmanteau test for white noise

-----

Portmanteau (Q) statistic = 23.5370

Prob > chi2(40) = 0.9822

\* We do not reject the null that the standardized squared residual are white noise, so the model capture the conditional heteroskedasticity.

# Testing if the model is stationary

test [ARCH]L1.arch + [ARCH]L2.arch + [ARCH]L1.garch = 1

( 1) [ARCH]L.arch + [ARCH]L2.arch + [ARCH]L.garch = 1

chi2( 1) = 8.38

Prob > chi2 = 0.0038

- We reject the null that the parameters add up to one (p-value is less than 1%) : the estimated model is stationary.

# VARIATIONS OF GARCH

- GARCH-t: the errors follow a t-student distribution;
- GARCH-M: the mean equation depends directly on the conditional standard deviation of the process (used to model weak correlation in asset returns).

Then, models that allows for asymmetric response to shocks: positive shocks have different effects on volatility than negative shocks. Bad news have a larger effects than good news. Leverage effect (volatility rise when returns fall and decline when returns rise)

- E-GARCH: Nelson (1991) – Exponential GARCH (does not require nonnegativity constraints.
- GJR-GARCH: Glosten, Jaganathan and Runkle (1994): decompose the effect of  $e_t^2$  into two effects using an interaction dummy variable.
- T-GARCH): Threshold GARCH – Shocks above certain threshold have different effect than below the threshold.

# ARCH-t

- The error  $u_t$  now are assuming to follow a t-distribution with  $v$  df (we can also assume the errors follow a skew normal; a skew t-distr.; a generalized error distr. (ged), for example).

$$\begin{aligned}Y_t &= \alpha_0 + \alpha_1 Y_{t-1} + e_t \\e_t &= \sigma_t u_t \\u_t &\sim t_v \\\sigma_t^2 &= \alpha_0 + \alpha_1 e_{t-1}^2 + \beta \sigma_{t-1}^2\end{aligned}$$

**STATA Command:**

**arch Y, arch(1) garch(1) distribution(t)**

# Leptokurtosis

```
tabstat X, statistics(kurtosis)
```

```
variable | kurtosis
```

```
-----+-----  
      X | 4.813832  
-----
```

- X is the %change in the US/UK exchange rate.
- The large kurtosis implies errors from a t-distribution.
- An GARCH-t(1,1) is estimated. Then we use the Ljung-Box test to check if the standardize residual and Standardize squared residuals are white-noise



# Ljung-Box test

Portmanteau test for white noise

-----

Portmanteau (Q) statistic = 4.3287

Prob > chi2(10) = 0.9313

wntestq e2V, lags (10)

Portmanteau test for white noise

-----

Portmanteau (Q) statistic = 8.5941

Prob > chi2(10) = 0.5710

- We do not reject the null. The GARCH-t(1,1) is a good model for the Dollar/Pound exchange rate.

# GARCH-M

- GARCH-M: GARCH-in-mean: Risk is a function of the conditional variance of the shock. In GARCH-M model, the variance is included in the mean equation:

$$Y_t = \alpha_0 + \alpha_1 \mathbf{X} + \lambda_1 \sigma_t + \lambda_2 \sigma_{t-1} + e_t$$

$$e_t = \sigma_t u_t$$

$$u_t \text{ is iid}(0,1)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 e_{t-1}^2 + \beta \sigma_{t-1}^2$$

**STATA Command:**

**arch y X, arch(1) garch(1) archm archmlags(1)**

\* archmlags is for additional lags of the variance on the mean equation.

# Predicting Excess Returns

- **Capital Asset Pricing Model:** a linear relationship between mean return and variance of an asset (we estimate the relative risk aversion). Our 3 variables are mean equity return (Dow Jones Industrial average; S&P 500, and NASDAQ) and the coefficient of the conditional standard deviation is the estimate of the relative risk aversion (high risk demands high return).
- The excess rates of return for each stock is computed (using the risk-free rate of return)
- Then a GARCH (1,1)-M(1) is estimated for these 3 excess returns.

# Predict Excess Returns in the US equity markets

## – GARCH-M model

|                                                                   | (1)<br>retDJIA           | (2)<br>retSP              | (3)<br>retNASDAQ         |                                      |
|-------------------------------------------------------------------|--------------------------|---------------------------|--------------------------|--------------------------------------|
| ----- t statistics in parentheses * p<0.10, ** p<0.05, *** p<0.01 |                          |                           |                          |                                      |
| main                                                              |                          |                           |                          |                                      |
| _cons                                                             | 0.0000800<br>(0.79)      | 0.000141<br>(1.55)        | 0.000336***<br>(3.50)    |                                      |
| -----                                                             |                          |                           |                          |                                      |
| ARCHM                                                             |                          |                           |                          |                                      |
| sigma2                                                            | 2.976**<br>(2.36)        | 2.633**<br>(2.23)         | 1.639*<br>(1.86)         | (relative risk aversion coefficient) |
| -----                                                             |                          |                           |                          |                                      |
| ARCH                                                              |                          |                           |                          |                                      |
| L.arch                                                            | 0.0827***<br>(47.74)     | 0.0885***<br>(45.78)      | 0.111***<br>(32.16)      |                                      |
| L.garch                                                           | 0.909***<br>(350.63)     | 0.907***<br>(361.21)      | 0.882***<br>(240.99)     |                                      |
| _cons                                                             | 0.00000100***<br>(11.05) | 0.000000755***<br>(10.23) | 0.00000130***<br>(13.18) |                                      |

# Asymmetric Response in GARCH: GJR-GARCH

- Unexpected event will have an asymmetric response. **Leverage effect:** volatility rise when returns fall and decline when returns rise, however positive shocks will have a smaller effect on volatility than a negative shock of the same magnitude.

$$\sigma_t^2 = \alpha_0 + \alpha_1 e_{t-1}^2 + \alpha_2 D_{t-1} e_{t-1}^2 + \beta \sigma_{t-1}^2$$

$$D_{t-1} = \begin{cases} 1 & \text{if } e < 0 \\ 0 & \text{otherwise} \end{cases}$$

**STATA Command:**

**arch y , arch(1) garch(1) tarch(1)**

\* archmlags is for additional lags of the variance on the mean equation.

# Asymmetric Response in GARCH: E-GARCH

- The Exponential GARCH assume there is a non-linear response to shocks (the variances are in log) and it substitute  $\alpha_1 e_{t-1}^2$  the variance function by a function  $g(z_{t-1})$ :

$$g(z_{t-1}) = \alpha_{11}z_{t-1} + \alpha_{12}(|z_{t-1}| - E|z_{t-1}|)$$

$$\text{Where } z_{t-1} = \frac{e_{t-1}}{\sigma_{t-1}} \quad (z_{t-1} \text{ is } N(0,1))$$

$$\Rightarrow E|z_{t-1}| = \sqrt{2/\pi}$$

$$g(z_{t-1}) = \alpha_{11}z_{t-1} + \alpha_{12}\left(|z_{t-1}| - \sqrt{2/\pi}\right)$$

$$\Rightarrow \ln(\sigma_t^2) = \alpha_0 + \alpha_{11}z_{t-1} + \alpha_{12}\left(|z_{t-1}| - \sqrt{2/\pi}\right) + \beta \ln(\sigma_{t-1}^2)$$

(the value of  $\sigma_t^2$  will not be negative)

# Asymmetric Response in GARCH: E-GARCH

- How the Exponential GARCH models asymmetric response to positive and negative shocks (how it allows for leverage effects?)

$$g(z_{t-1}) = \begin{cases} (\alpha_{11} + \alpha_{12})z_{t-1} - \alpha_{12}\sqrt{2/\pi} & \text{if } e_{t-1} \geq 0 \\ (-\alpha_{11} + \alpha_{12})z_{t-1} - \alpha_{12}\sqrt{2/\pi}, & \text{otherwise} \end{cases}$$

STATA Command:

`arch y , earch(1/2) egarch(1/3)`

# How to test for Leverage Effects

- Engle and Ng (1993) developed a test to determine if there is a leverage effect: the Sign Bias Test:
- Let  $\{s_t\}$  be a series of each residual divided by its standard deviation. Then estimate the following equation:

$$s_t^2 = a_0 + a_1 D_{t-1} + a_2 D_{t-1} s_{t-1} + a_3 (1 - D_{t-1}) + e_t$$

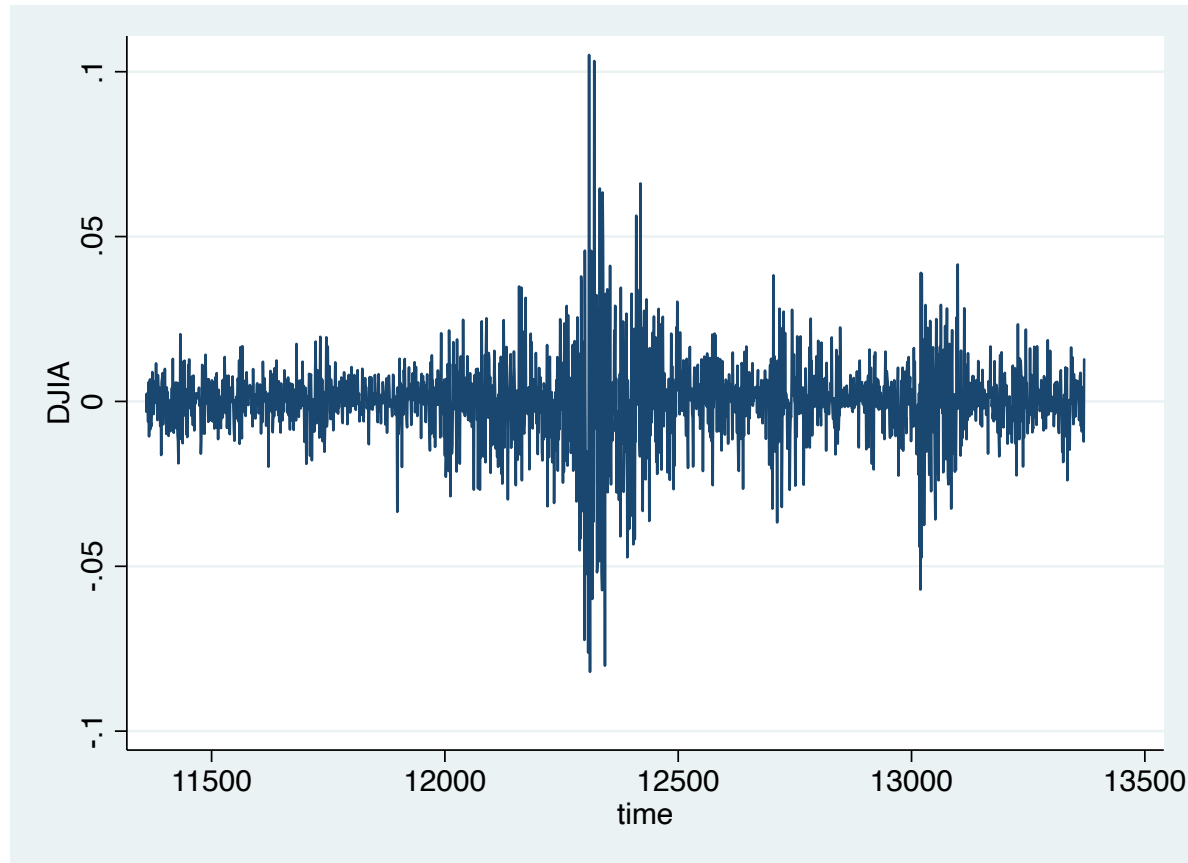
Where:  $D_{t-1} = \begin{cases} 1 & \text{if } e < 0 \\ 0, & \text{otherwise} \end{cases}$  and  $e_t$  is a regression residual

\* For  $a_1$  different than zero, means that the sign of the current period shock is helpful in predicting the conditional volatility.  $a_2$  and  $a_3$ : test if positive and negative shocks depend on their size.



# Comparing Asymmetric GARCH models

Dow Jones Industrial Average %Change



# Comparing Asymmetric GARCH models

```
. esttab GARCH GJR_GARCH E_GARCH, not
```

|           | (1)<br>DJIA | (2)<br>DJIA | (3)<br>DJIA |
|-----------|-------------|-------------|-------------|
| DJIA      |             |             |             |
| _cons     | 0.000476    | 0.000411    | 0.000298    |
| ARCH      |             |             |             |
| L.arch    | 0.278***    | 0.314***    |             |
| L.garch   | 0.828***    | 0.871***    |             |
| L.tarch   |             | -0.0825     |             |
| L.earch   |             |             | -0.0830***  |
| L.earch_a |             |             | 0.437***    |
| L.egarch  |             |             | 1.207***    |
| _cons     | -0.0000182  | -0.0000220* | 1.770*      |
| N         | 1579        | 1579        | 1579        |

\* p<0.05, \*\* p<0.01, \*\*\* p<0.001

```
. estimates stats GARCH GJR_GARCH E_GARCH
```

| Model     | Obs  | ll(null) | ll(model) | df | AIC       | BIC       |
|-----------|------|----------|-----------|----|-----------|-----------|
| GARCH     | 1579 | .        | 4823.805  | 4  | -9639.611 | -9618.152 |
| GJR_GARCH | 1579 | .        | 4824.555  | 5  | -9639.111 | -9612.288 |
| E_GARCH   | 1579 | .        | 4825.669  | 5  | -9641.337 | -9614.515 |

Note: N=Obs used in calculating BIC; see [R] BIC note

```
. correlate GARCH_hat E_GARCH_hat GJR_GARCH_hat
(obs=2013)
```

|               | GARCH_~t | E_GARCH_~t | GJR_GARCH_~t |
|---------------|----------|------------|--------------|
| GARCH_hat     | 1.0000   |            |              |
| E_GARCH_hat   | 0.8967   | 1.0000     |              |
| GJR_GARCH_hat | 0.9908   | 0.9262     | 1.0000       |

# IGARCH: Integrated GARCH

Due to the fact that

$$\sum_{i=1}^p \alpha_i + \sum_{i=1}^q \beta_i = 1$$

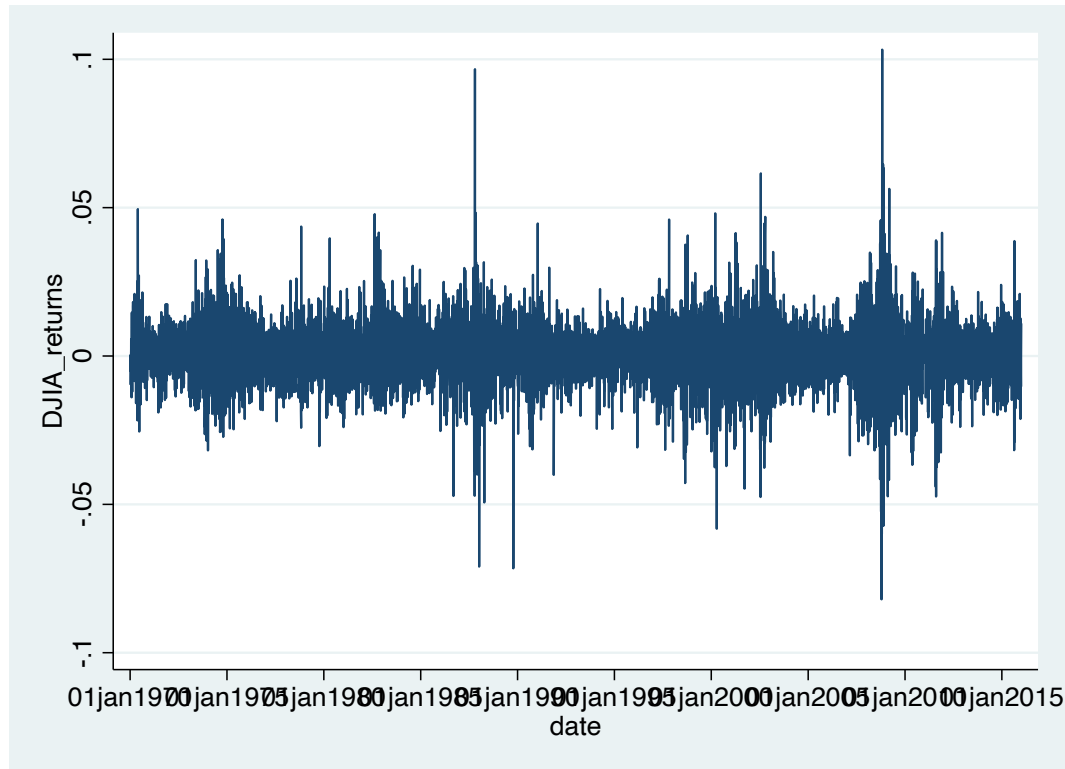
There is no finite conditional variance, so the unconditional variance is not mean-reverting. The effect of a shock is permanent.

The IGARCH allows for infinite variances and leptokurtosis (positive excessive kurtosis), characteristics of many financial time series.

Unlike a GARCH, the IGARCH conditional variance will be a geometrically decaying function of the current and past values of the  $\{e_t^2\}$  and can be estimated any GARCH, by imposing the constraint that  $\sum_{i=1}^p \alpha_i + \sum_{i=1}^q \beta_i = 1$

# Dow Jones Industrial Average daily returns

## %Change: 1970-2015



# IGARCH: example (Down Jones returns)

```
constraint 1 [ARCH]_b[L1.arch] + [ARCH]_b[L1.garch] = 1
```

```
arch DJIA_returns, arch(1) garch(1) nolog constraints(1)
```

```
-----  
DJIA_returns |    Coef.  Std. Err.    z   P>|z|   [95% Conf. Interval]  
-----+-----  
DJIA_returns |  
  _cons |   .0004095   .0000938   4.36  0.000   .0002256   .0005933  
-----+-----  
ARCH         |  
  arch |  
    L1. |   .1841262   .0089848  20.49  0.000   .1665165   .201736  
      |  
  garch |  
    L1. |   .8158738   .0089848  90.81  0.000   .798264   .8334835  
      |  
  _cons |  -8.42e-08   4.74e-07   -0.18  0.859  -1.01e-06   8.45e-07  
-----
```

```
.display [ARCH]_b[L1.arch] + [ARCH]_b[L1.garch]
```

```
1
```